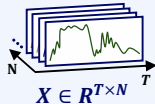


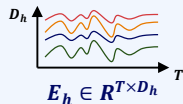
Input Interface

Multi-modal Inputs

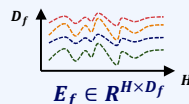
Historical Target Status



Historical Exogenous (ExG_h)



Future Exogenous (ExG_f)



Graph Metadata (M_G)

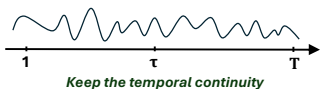


Adjacency / Coordinates / Attributes

AGSP Module

Adaptive Gaussian-soft-windowed Patching

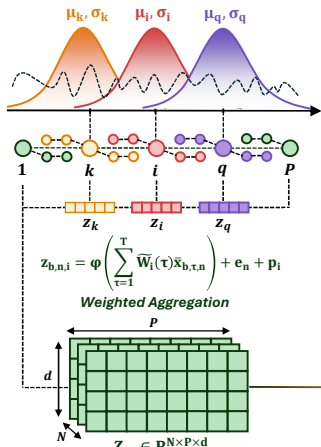
Continuous Time Axis



Soft Overlapping Gaussian Windows

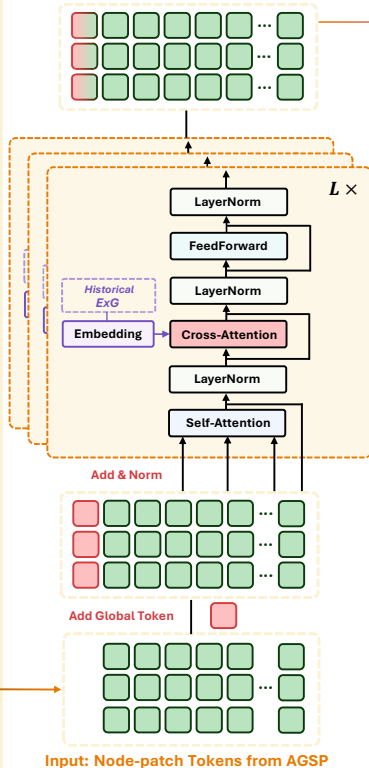
$$W_i(\tau) = \exp\left(-\frac{(\tau - \mu_i)^2}{2\sigma_i^2 + \varepsilon}\right)$$

$$\tilde{W}_i(\tau) = \frac{W_i(\tau)}{\sum_{j=1}^p W_j(\tau) + \varepsilon}$$



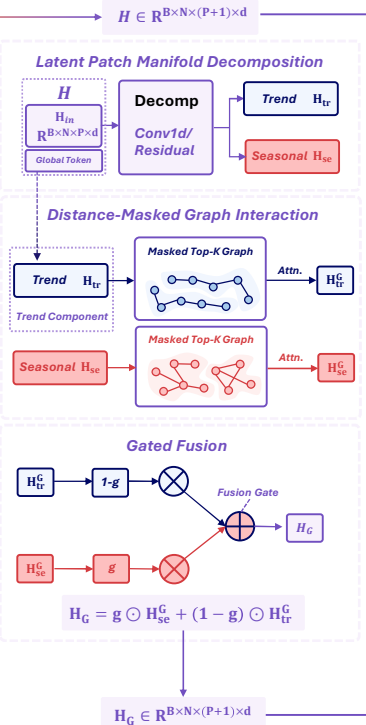
Temporal Transformer

Multi-head Self-Attention with Global Token



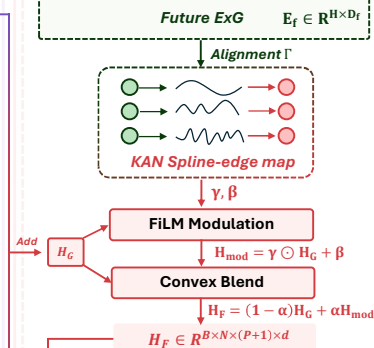
R-STMF Module

Relational Spatio-Temporal Manifold Fusion



K-AMC Module

KAN-based Exogenous Modulation



PI-DCH Module

Physics-Informed Dynamic Capping

